

## 1. General Description

The EMS4002 is a bidirectional low-power dual port, high-speed, USB 2.0 analog switch with integrated protection for USB Type-C™ systems. The device is configured as a dual 2:1 or 1:2 switch. It is optimized for use with the USB 2.0 DP/DM lines in a USB Type-C™ system.

The device integrates over-voltage protection on the C0+/- pins can withstand up to DC 7V with automatic shutoff circuitry in order to protect system components behind the switch. GPIO controls of SEL and  $\overline{\text{EN}}$  are 1.8V logic compatible.

The EMS4002 is available in QFN 1.4x1.8-10L, QFN 1.5x2.0-10L, DFN3x3-10L and MSOP-10L package with Pb-free and Halogen-free making it a perfect candidate for mobile and space constrained applications.

## 2. Features and Benefits

- Wide Supply Range from 2.5 V to 4.5 V
- Differential 2:1 or 1:2 Switch/Multiplexer
- Up to DC 7V Overvoltage Protection (OVP) on C0+/- Ports
- IEC 64000-4-5 Surge Protection w/o External TVS onto C0+/- Ports:  $\pm 20$  V
- System Side Clamp Voltage Pulse Less than 8 V, Duration Less than 450 ns
- Power off Protection When  $V_{DD} = 0$  V
- 8.5  $\Omega$  typical ON resistance
- Insertion loss: -1dB@580MHz, -2dB@1GHz, -3dB@1.3GHz
- 4 pF typical ON capacitance
- 1.8 V Compatible Control Logic Inputs
- Latch-up performance exceeds 250 mA
- ESD protection:
  - HBM ESDA/JEDEC JS-001 Class 2 exceeds 2000 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V

## 3. Applications

- USB Type-C™, Type A and Micro B devices
- Mobile Phones, Tablets and Notebooks

## 4. Ordering Information

Table 1. Ordering information

| Type number | Topside marking | Package        |                                                               |          |
|-------------|-----------------|----------------|---------------------------------------------------------------|----------|
|             |                 | Name           | Description                                                   | Quantity |
| EMS4002RSW  | 4002YYWW        | QFN1.4×1.8-10L | QFN package, 10 pins<br>1.4 mm × 1.8 mm; 0.55 mm (Max) height | 3000     |
| EMS4002RSE  | 4002YYWW        | QFN1.5×2.0-10L | QFN package, 10 pins<br>1.5 mm × 2.0 mm; 0.6 mm (Max) height  | 3000     |
| EMS4002DRC  | 4002YYWW        | DFN3x3-10L     | DFN package, 10 pins<br>3.0 mm × 3.0 mm; 0.8 mm (Max) height  | 5000     |
| EMS4002MS   | 4002YYWW        | MSOP-10L       | MSOP package, 10 pins<br>3.0mm × 3.0mm; 1.1 mm (Max) height   | 3000     |

### MARKING INFORMATION

NOTE: YYWW = Data Code.

YY WW



## 5.Functional Diagram

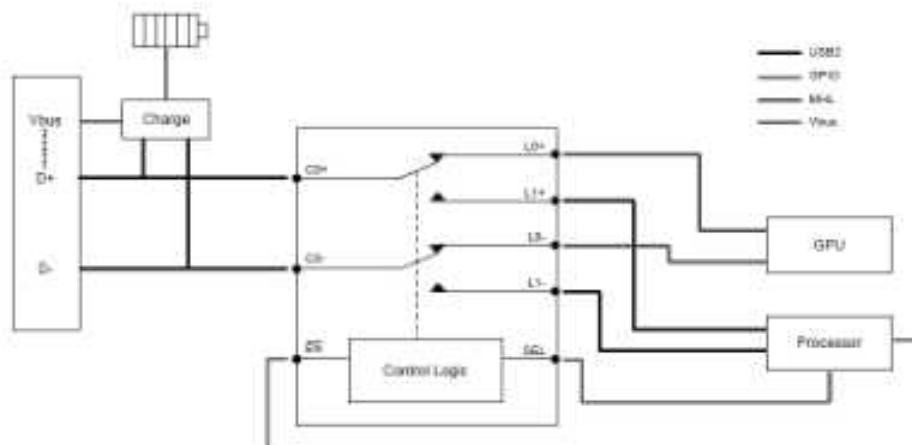


Fig 1. Functional diagram

## 6. Pinning Information

### 6.1. Pin map

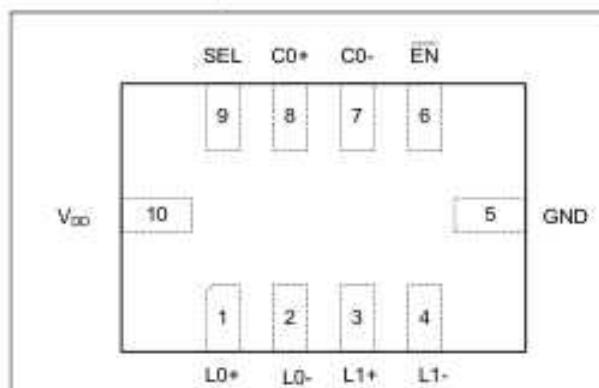


Fig 2. Top view pin configuration QFN 1.5x2.0-10L

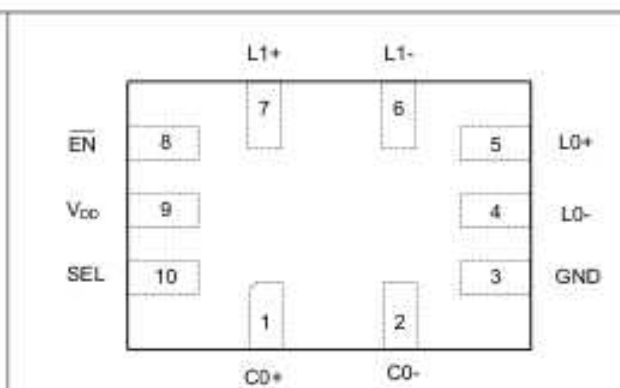


Fig 3. Top view pin configuration QFN 1.4x1.8-10L

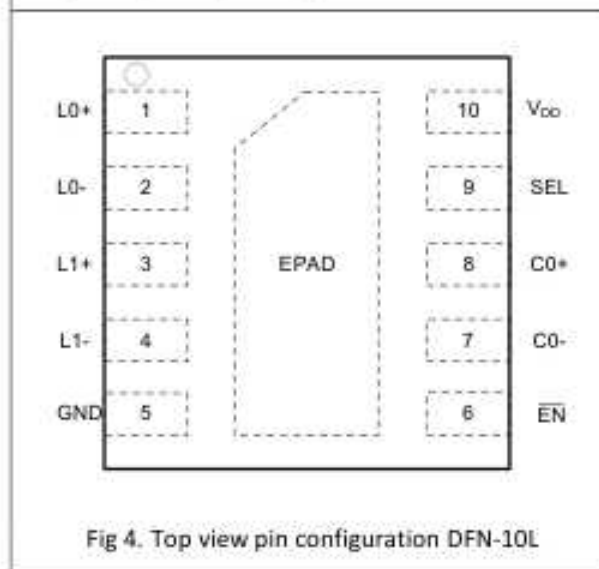


Fig 4. Top view pin configuration DFN-10L

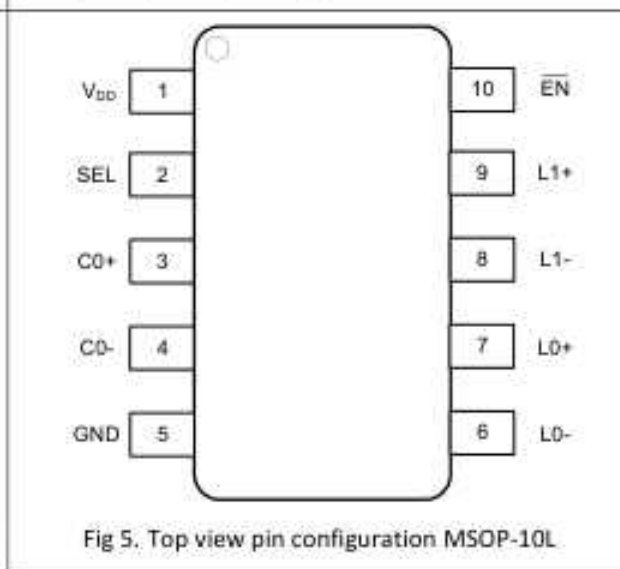


Fig 5. Top view pin configuration MSOP-10L

## 6.2. Pin description

Table 2. Pin description

| Symbol          | Pin            |                           |          | Signal Type | Description                                                         |
|-----------------|----------------|---------------------------|----------|-------------|---------------------------------------------------------------------|
|                 | QFN1.4x1.8-10L | QFN1.5x2.0-10L/DFN3x3-10L | MSOP-10L |             |                                                                     |
| C0+             | 1              | 8                         | 3        | I/O         | Signal I/O, Common Port                                             |
| C0-             | 2              | 7                         | 4        | I/O         | Signal I/O, Common Port                                             |
| L1+             | 7              | 3                         | 9        | I/O         | Signal I/O, Channle 1                                               |
| L1-             | 6              | 4                         | 8        | I/O         | Signal I/O, Channle 1                                               |
| L0+             | 5              | 1                         | 7        | I/O         | Signal I/O, Channle 0                                               |
| L0-             | 4              | 2                         | 6        | I/O         | Signal I/O, Channle 0                                               |
| SEL             | 10             | 9                         | 2        | I           | Operation Model Select;<br>(when SEL=0: C0->L0, when SEL=1: C0->L1) |
| EN              | 8              | 6                         | 10       | I           | EN = 1, Power Down is Enabled                                       |
| V <sub>DD</sub> | 9              | 10                        | 1        | PWR         | Positive Supply Voltage                                             |
| GND             | 3              | 5                         | 5        | GND         | Power Ground                                                        |

## 7.Functional Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level; X = Don't care.

| Input |     | Switch                 |
|-------|-----|------------------------|
| EN    | SEL |                        |
| L     | L   | C0+ to L0+, C0- to L0- |
| L     | H   | C0+ to L1+, C0- to L1- |
| H     | X   | Switch off             |

## 8.Absolute Maximum Ratings

Absolute Maximum Ratings may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification is not implied.

**Table 4 Absolute Maximum Ratings**

| Symbol           | Parameter                                | Conditions | Min  | Max | Unit |
|------------------|------------------------------------------|------------|------|-----|------|
| V <sub>DD</sub>  | supply voltage                           |            | -0.3 | 5.5 | V    |
| V <sub>IN</sub>  | control input voltage                    |            | -0.3 | 5.5 | V    |
| P <sub>D</sub>   | power dissipation                        |            |      | 250 | mW   |
| T <sub>stg</sub> | storage temperature                      |            | -55  | 150 | °C   |
| T <sub>J</sub>   | junction temperature under bias          |            |      | 125 | °C   |
| T <sub>L</sub>   | lead temperature (soldering, 10 seconds) |            |      | 260 | °C   |

## 9.Recommended Operating Conditions

**Table 5 Recommend operating ratings**

| Symbol          | Parameter                | Conditions | Min | Max | Unit |
|-----------------|--------------------------|------------|-----|-----|------|
| V <sub>DD</sub> | supply voltage operating |            | 2.5 | 4.5 | V    |
| V <sub>IN</sub> | control input voltage    |            | 0   | 4.5 | V    |
| T <sub>A</sub>  | operating temperature    |            | -40 | 85  | °C   |

## 10. Electrical Characteristics

**Table 4. Electrical characteristics**
 $T_A = 25\text{ }^{\circ}\text{C}$ ,  $V_{DD} = 3.3\text{ V}$ , unless otherwise specified

| Symbol                               | Parameter                                     | Conditions                                                                                  | Min | Typ | Max | Unit |
|--------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------|-----|-----|-----|------|
| POWER SUPPLY                         |                                               |                                                                                             |     |     |     |      |
| V <sub>DD</sub>                      | supply voltage range                          |                                                                                             | 2.5 | 3.3 | 4.5 | V    |
| I <sub>DD</sub>                      | supply current                                | $\overline{EN} = 3.5\text{ V}$ $V_{DD} = 3.5\text{ V}$                                      |     | 0.3 | 2   | uA   |
|                                      |                                               | $\overline{EN} = 0\text{ V}$ $V_{DD} = 3.5\text{ V}$                                        |     | 20  | 40  | uA   |
| SEL/_EN DIGITAL INPUT CONTROL        |                                               |                                                                                             |     |     |     |      |
| V <sub>IH</sub>                      | control input logic high                      |                                                                                             | 1.6 |     | 4.5 | V    |
| V <sub>IL</sub>                      | control input logic low                       |                                                                                             | 0   |     | 0.5 | V    |
| SWITCH ON RESISTANCE AND OFF LEAKAGE |                                               |                                                                                             |     |     |     |      |
| R <sub>ON</sub>                      | on-resistance                                 | V <sub>IS</sub> = 0 V to 0.4 V I <sub>OUT</sub> = 8 mA                                      |     | 8.5 | 10  | Ω    |
| R <sub>FLAT</sub>                    | R <sub>ON</sub> flatness [1]                  | V <sub>IS</sub> = 0 V to 0.4 V I <sub>OUT</sub> = 8 mA                                      |     | 0.1 | 0.5 | Ω    |
| ΔR <sub>ON</sub>                     | R <sub>ON</sub> matching between channels [2] | V <sub>IS</sub> = 0 V to 0.4 V I <sub>OUT</sub> = 8 mA                                      |     | 0.1 | 0.5 | Ω    |
| I <sub>LEAK</sub>                    | OFF leakage current                           | V <sub>CD+/-</sub> = 6 V V <sub>LD+/-</sub> = V <sub>LS+/-</sub> = 0 V                      |     | 3.5 | 6   | uA   |
| SWITCH DYNAMICS                      |                                               |                                                                                             |     |     |     |      |
| C <sub>ON</sub>                      | on capacitance                                | V <sub>DD</sub> = V <sub>IS</sub> = 3.3 V, $\overline{EN} = 0$                              |     | 4   |     | pF   |
| C <sub>OFF</sub>                     | off capacitance                               | V <sub>DD</sub> = V <sub>IS</sub> = 3.3 V, $\overline{EN} = 1$                              |     | 3.5 |     | pF   |
| Off                                  | off isolation                                 | f = 250 MHz, R <sub>L</sub> = 50 Ω, C <sub>L</sub> = 0 pF                                   |     | -38 |     | dB   |
| X <sub>TALK</sub>                    | crosstalk (Channel-to-Channel) [3]            | f = 250 MHz, R <sub>L</sub> = 50 Ω, C <sub>L</sub> = 0 pF                                   |     | -41 |     | dB   |
| BW                                   | -3dB bandwidth                                | R <sub>L</sub> = 50 Ω, C <sub>L</sub> = 0 pF<br>Signal Power = 0 dBm                        |     | 1.3 |     | GHz  |
| BBM                                  | break-before-make                             | V <sub>IS</sub> = 0.4 V, R <sub>L</sub> = 50 Ω                                              |     | 640 |     | ns   |
| t <sub>ON</sub>                      | turn-on time                                  | V <sub>IS</sub> = 0.4 V, R <sub>L</sub> = 50 Ω<br>$\overline{EN}$ switches from High to Low |     | 45  |     | us   |
| t <sub>OFF</sub>                     | turn-off time                                 | V <sub>IS</sub> = 0.4 V, R <sub>L</sub> = 50 Ω<br>$\overline{EN}$ switches from Low to High |     | 130 |     | ns   |
| t <sub>PD</sub>                      | propagation delay                             | V <sub>IS</sub> = 0.4 V, R <sub>L</sub> = 50 Ω                                              |     | 400 |     | ps   |

## EMS4002

### Dual 2:1 USB2.0 Mux/De-Mux with 20V Over-Voltage Protection

| Symbol                         | Parameter                        | Conditions                                         | Min | Typ | Max | Unit |
|--------------------------------|----------------------------------|----------------------------------------------------|-----|-----|-----|------|
| <b>OVER VOLTAGE PROTECTION</b> |                                  |                                                    |     |     |     |      |
| $V_{FP}$                       | OVP lockout threshold            | $V_{C0+/-}$ Rising Edge                            | 4.6 | 4.9 | 5.2 | V    |
| $V_{HYS}$                      | OVP hysteresis                   |                                                    |     | 300 |     | mV   |
| $V_{CLAMP}$                    | clamp voltage on L0+/- and L1+/- | 10 V shorts to C0+/- with $R_L = 1\text{ k}\Omega$ |     |     | 8   | V    |
| $t_{FP}$                       | OVP response time                | 10 V shorts to C0+/- with $R_L = 1\text{ k}\Omega$ |     | 330 | 450 | ns   |
| $t_{FPR}$                      | OVP recovery time                | $V_{C0+/-}$ from 6 V to 1 V                        |     | 100 |     | ns   |

[1] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.

[2]  $R_{ON}$  matching between channels is calculated by subtracting the channel with the lowest max Ron value from the channel with the highest max  $R_{ON}$  value.

[3] Crosstalk is inversely proportional to source impedance.

## 11. Typical Performance Curves

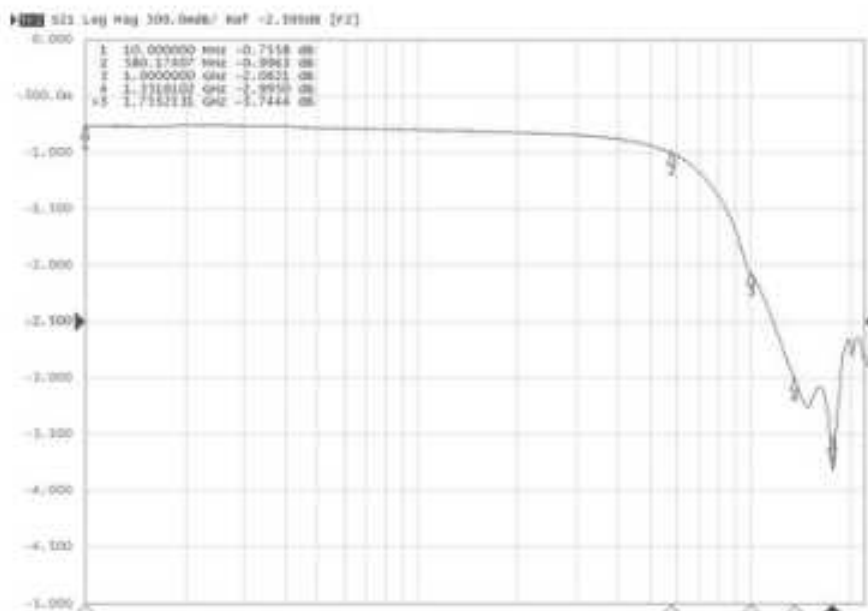


Fig 6. Switch Bandwidth or Insertion Loss

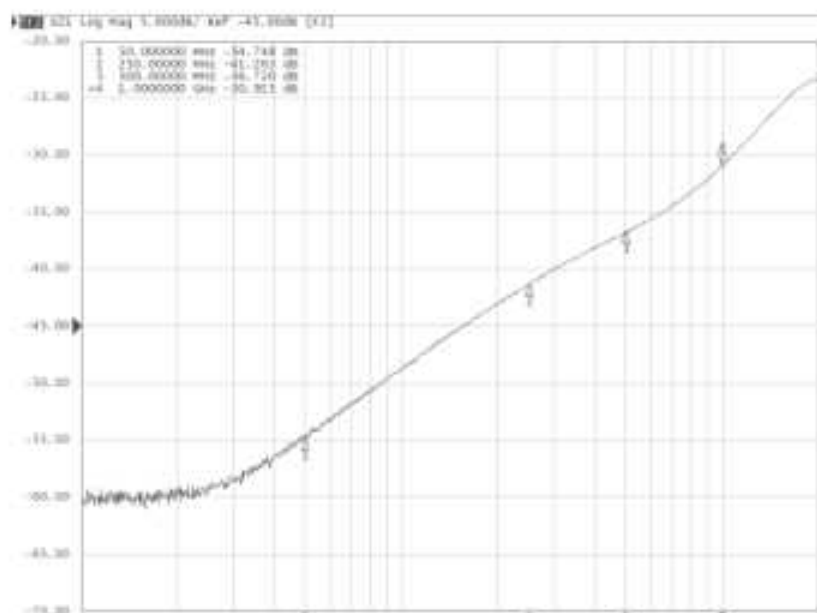


Fig 7. Switch Channel to Channel Cross-Talk

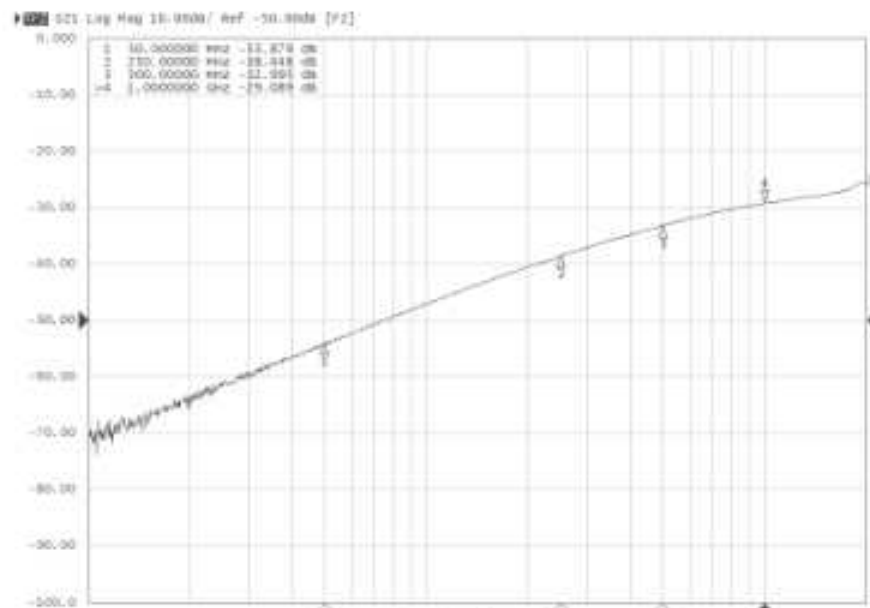


Fig 8. Switch Off Isolation

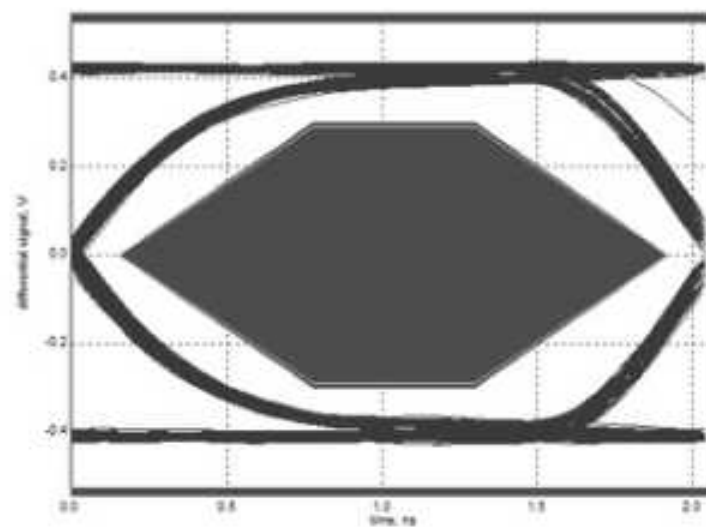
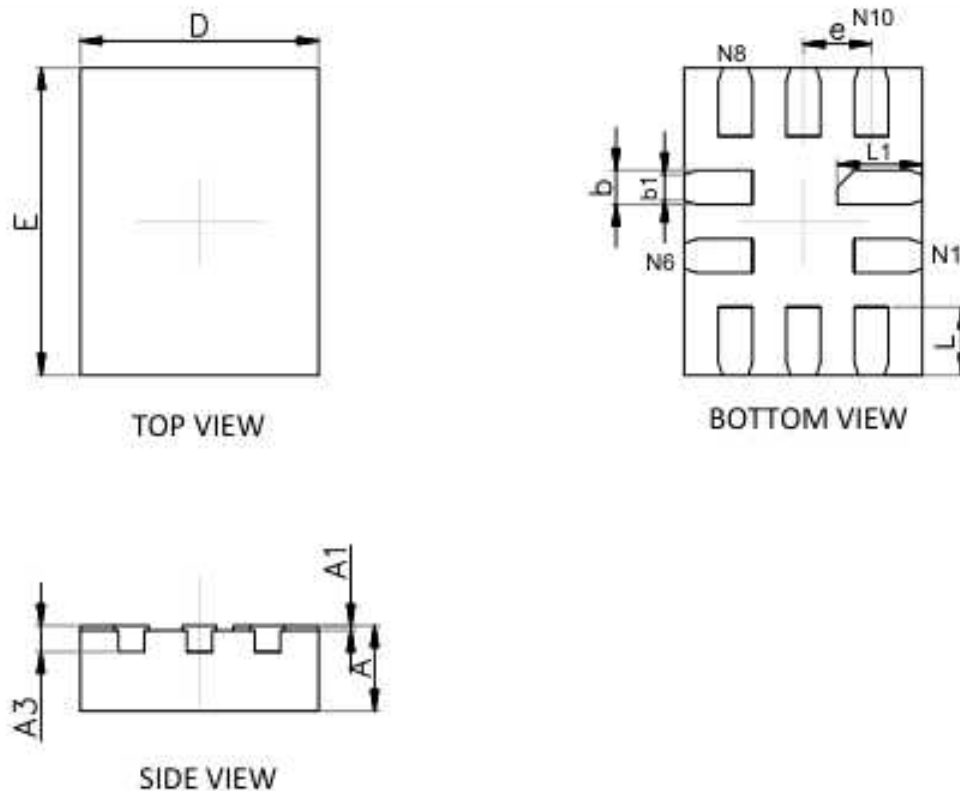


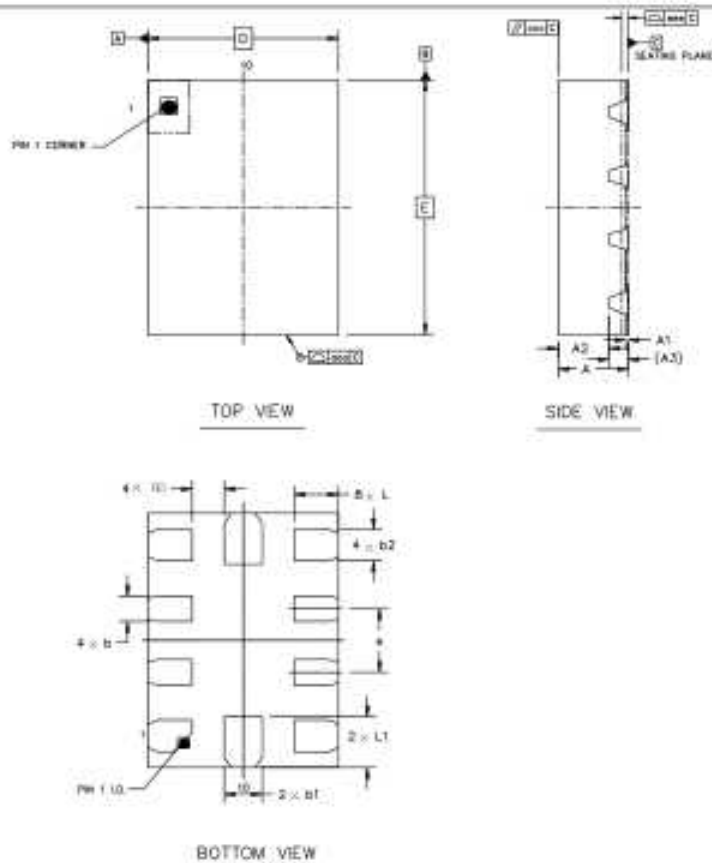
Fig 9. Eye Pattern

## 12. Package Outline

QFN 1.4x1.8 -10L



| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min.                      | Max.  | Min.                 | Max.  |
| A      | 0.450                     | 0.550 | 0.018                | 0.022 |
| A1     | 0.000                     | 0.050 | 0.000                | 0.002 |
| A3     | 0.152REF.                 |       | 0.006REF.            |       |
| D      | 1.350                     | 1.450 | 0.053                | 0.057 |
| E      | 1.750                     | 1.850 | 0.069                | 0.073 |
| b      | 0.150                     | 0.250 | 0.006                | 0.010 |
| b1     | 0.100                     | 0.200 | 0.004                | 0.008 |
| e      | 0.400TYP.                 |       | 0.016TYP.            |       |
| L      | 0.350                     | 0.450 | 0.014                | 0.018 |
| L1     | 0.450                     | 0.550 | 0.018                | 0.022 |

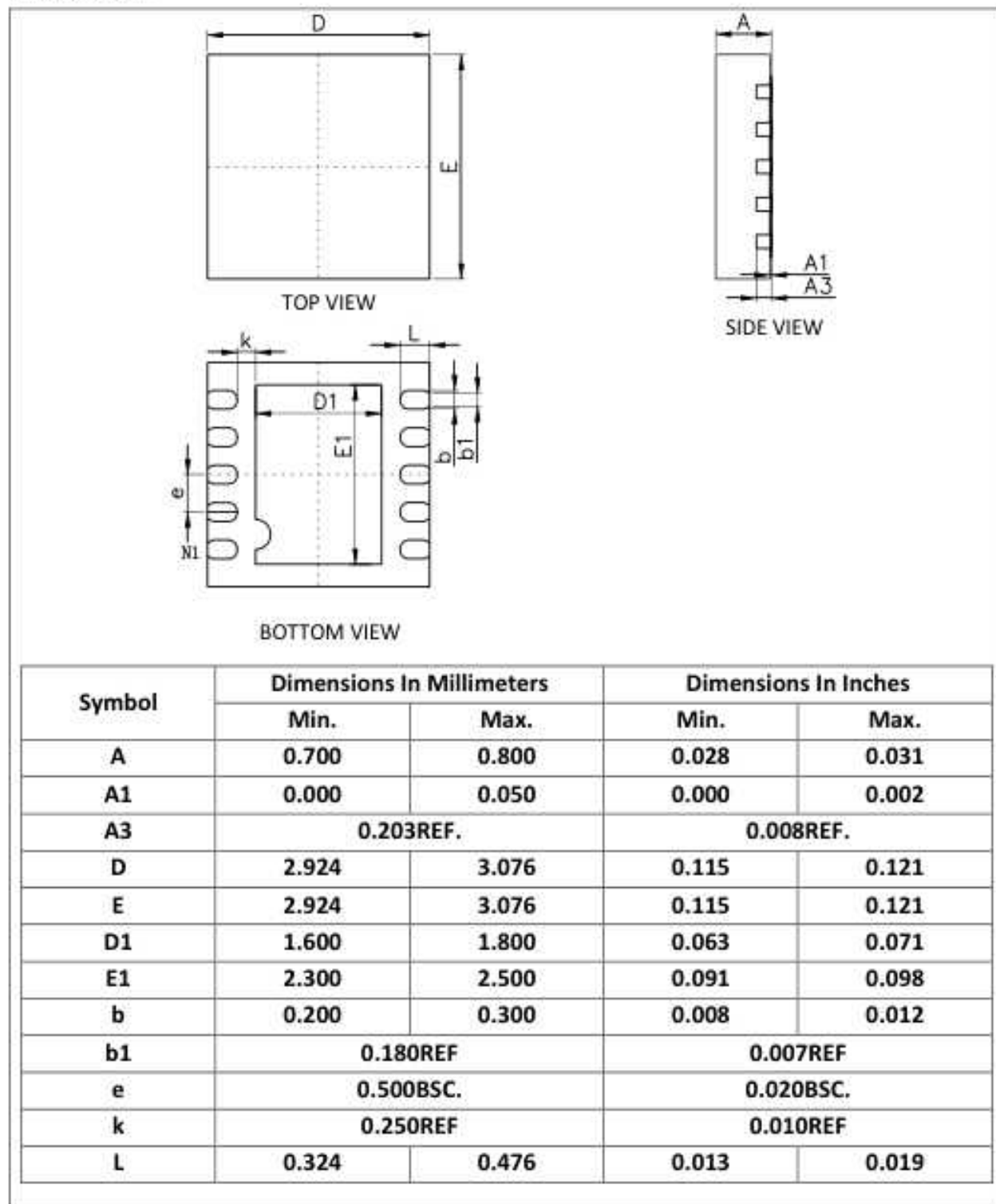
**QFN 1.5x2.0 -10L**


| Symbol | Dimensions In Millimeters |       |      |
|--------|---------------------------|-------|------|
|        | MIN                       | NOM   | MAX  |
| A      | 0.5                       | 0.55  | 0.6  |
| A1     | 0                         | 0.02  | 0.05 |
| A2     | ---                       | 0.398 | ---  |
| A3     | 0.152 REF                 |       |      |
| D      | 1.5 BSC                   |       |      |
| E      | 2 BSC                     |       |      |
| b      | 0.15                      | 0.2   | 0.25 |
| b1     | 0.25                      | 0.3   | 0.35 |
| b2     | 0.2                       | 0.25  | 0.3  |
| e      | 0.5 BSC                   |       |      |
| L      | 0.3                       | 0.35  | 0.4  |
| L1     | 0.35                      | 0.40  | 0.45 |
| K      | 0.25 ref                  |       |      |

# EMS4002

## Dual 2:1 USB2.0 Mux/De-Mux with 20V Over-Voltage Protection

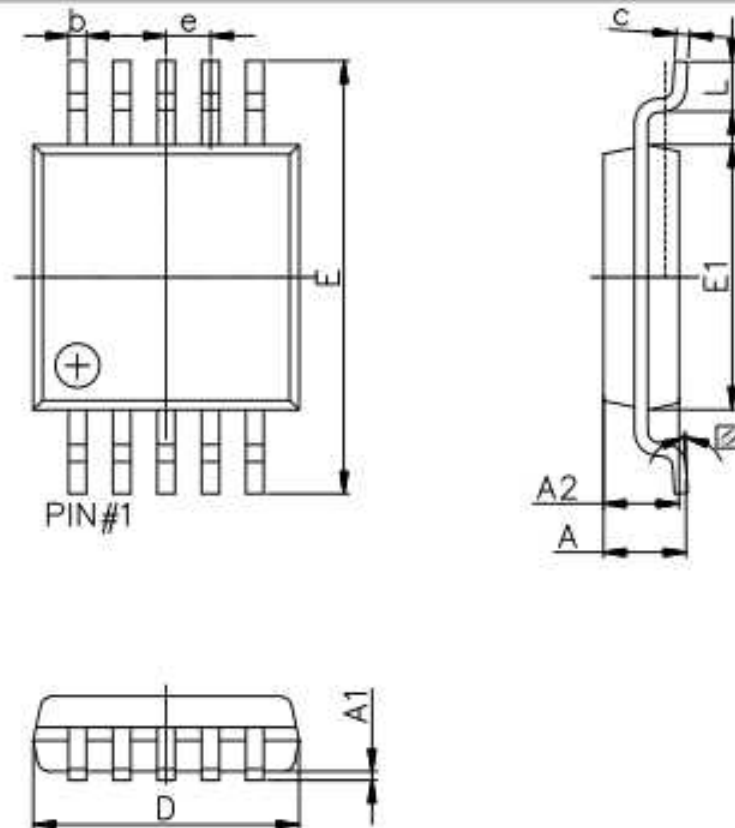
### DFN 3x3 -10L



# EMS4002

## Dual 2:1 USB2.0 Mux/De-Mux with 20V Over-Voltage Protection

### MSOP - 10L



| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min.                      | Max.  | Min.                 | Max.  |
| A      | ---                       | 1.100 | ---                  | 0.043 |
| A1     | 0.020                     | 0.150 | 0.001                | 0.006 |
| A2     | 0.750                     | 0.950 | 0.030                | 0.037 |
| b      | 0.180                     | 0.330 | 0.007                | 0.013 |
| c      | 0.090                     | 0.230 | 0.004                | 0.009 |
| D      | 2.900                     | 3.100 | 0.114                | 0.122 |
| e      | 0.500(BSC)                |       | 0.020(BSC)           |       |
| E      | 4.750                     | 5.050 | 0.187                | 0.199 |
| E1     | 2.900                     | 3.100 | 0.114                | 0.122 |
| L      | 0.400                     | 0.800 | 0.016                | 0.031 |
| θ      | 0°                        | 6°    | 0°                   | 6°    |

## 13. Tape and Reel Information

### 13.1. Carrier tape dimensions

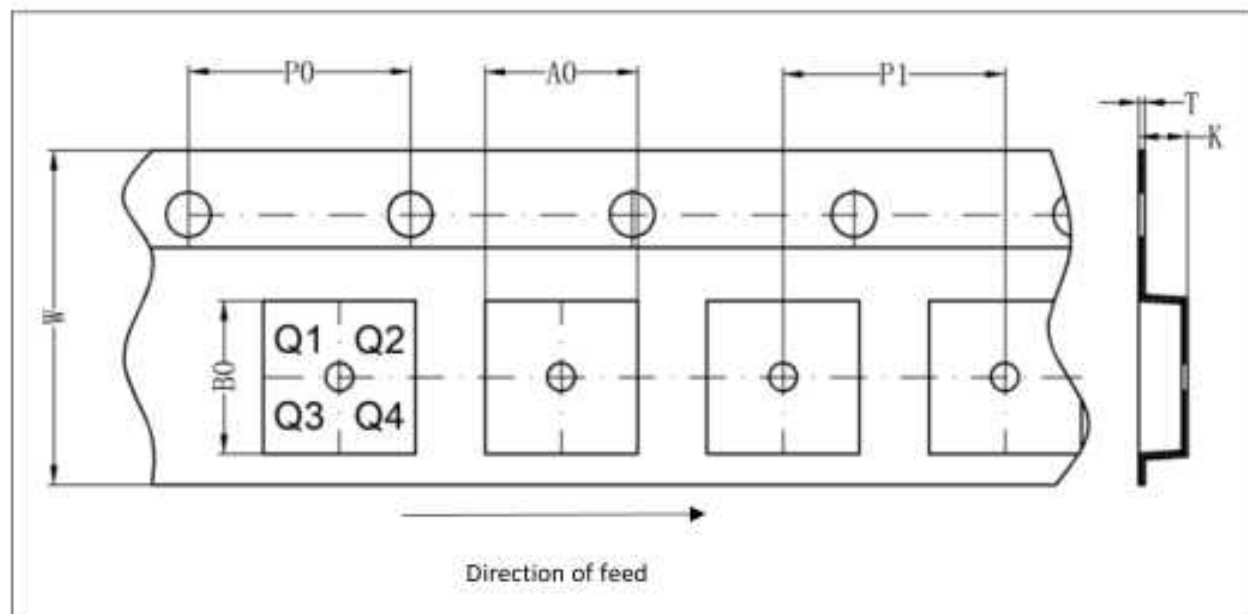
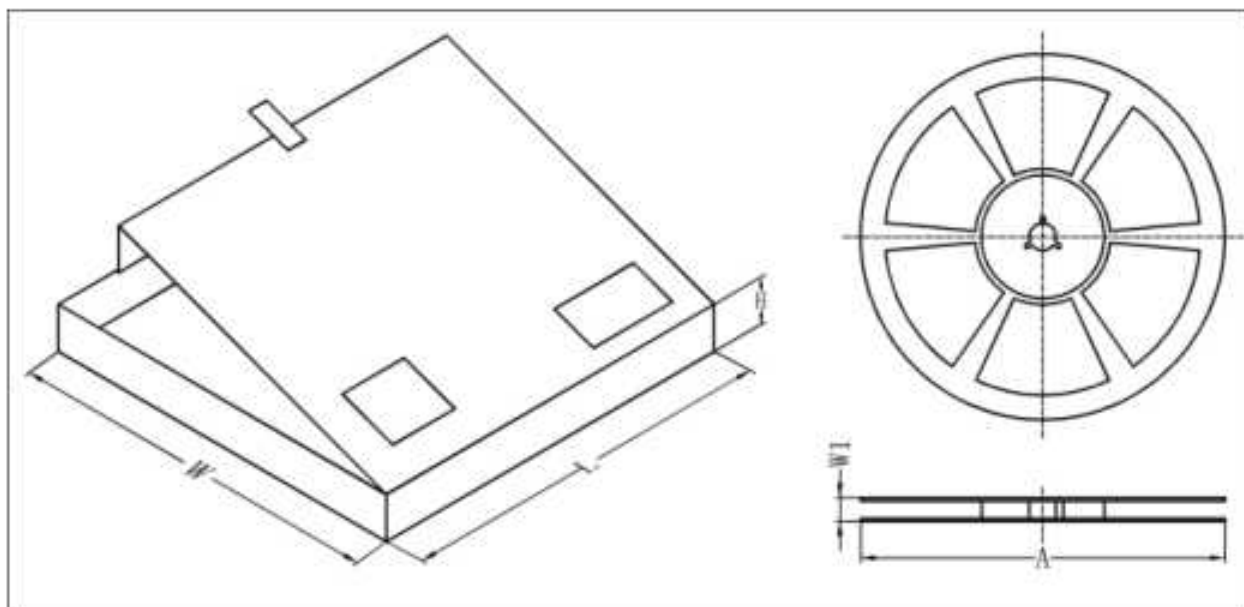


Table 5. Carrier tape dimensions

| Package version | A0(mm) | B0(mm) | K0(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|--------|-------|--------|-------|--------|-------|
| QFN1.4×1.8-10L  | 1.6    | 2.0    | 0.85   | 0.2   | 4      | 8     | 4      | Q1    |
| DFN3x3-10L      | 3.35   | 3.35   | 1.13   | 0.25  | 8      | 12    | 4      | Q1    |
| QFN1.5x2-10L    | 1.7    | 2.3    | 0.75   | 0.2   | 4      | 8     | 4      | Q1    |
| MSOP-10L        | 5.3    | 3.4    | 1.4    | 0.25  | 8      | 12    | 4      | Q1    |

## 13.2. Reel and box dimensions



**Table 6. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Max Reel Width W1 (mm) | SPQ (pcs)[1] | Reels per box | Outer box dimensions LxWxH(mm)[2] |
|-----------------|-----------------|-----------------------|------------------------|--------------|---------------|-----------------------------------|
| QFN1.4x1.8-10L  | RSW             | 180                   | 13.2                   | 3000         | 1             | 210x200x40                        |
| DFN3x3-10L      | DRC             | 330                   | 18.4                   | 5000         | 1             | 358x340x50                        |
| QFN1.5x2-10L    | RSE             | 180                   | 13.2                   | 3000         | 1             | 210x200x40                        |
| MSOP-10L        | MS              | 330                   | 18.4                   | 3000         | 1             | 358x340x50                        |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 14. Abbreviations

Table 7. Abbreviations

| Acronym | Description             |
|---------|-------------------------|
| ESD     | ElectroStatic Discharge |
| HBM     | Human Body Model        |

## 15. Revision History

Table 8. Revision history

| Document ID      | Release Date | Data sheet status | Change notice | Supersedes |
|------------------|--------------|-------------------|---------------|------------|
| EMS4002 Rev. 1.0 | Feb 10, 2026 | Product datasheet |               |            |